

Hands-on Lab: Create Tables and Load Data in Datasette



Estimated time needed: 20 minutes

In this lab, you will learn how to create tables and load data in Datasette.

Objectives

After completing this lab, you will be able to:

- Create and load data into a table from a CSV file
- Create and load data into a table from a SQL script file

Prerequisites

In this lab, you will use [Datasette](#), an open-source multi-tool for exploring and publishing data.

Datasets

PETSHOP and BookShop are the two data sets you will use in this lab.

- PETSHOP:

ID	ANIMAL
1	Cat
2	Dog
3	Parrot
4	Hamster
5	Goldfish

- BookShop:

BOOK_ID	TITLE	AUTH
B101	Introduction to Algorithms	Thom
B201	Structure and Interpretation of Computer Pro...	Harol
B301	Deep Learning	Ian G
B401	Algorithms Unlocked	Thom
B501	Machine Learning: A Probabilistic Perspective	Kevin

Exercise 1: Load a CSV file and create a table using the Datasette tool

In this exercise, you will learn how to load a CSV file and create a table using the Datasette tool.

1. First, select **Open tool**, then click the **Navigation Pane** at the right-end corner, and then select **Add DataSets**.

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Add a Dataset

To add a dataset into your lab, insert the link to the full URL to the CSV dataset below.

Full URL to Dataset:

Create

Powered by [Datasette](#)

2. You will then be redirected to a page where you need to enter the full URL of the CSV data set in the text box.
 - Right-click the link [PETSHOP.csv](#) and copy the link address. Enter the copied URL in the text box and select the **create** button.

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Add a Dataset

To add a dataset into your lab, insert the link to the full URL to the CSV dataset below.

Full URL to Dataset:

Create

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- The data loaded from the CSV file will create the PETSHOP table. By default, a **SELECT** query related to the table will appear on the **text area** section of the following webpage. Click **Submit Query** to view the results.

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Database: PETSHOP

```
SELECT * FROM PETSHOP;
```

Tip: Autocomplete with Ctrl+Enter or Cmd+Enter

Submit query

Results

All commands ran successfully

```
SELECT * FROM PETSHOP
```

ID	ANIMAL	SALEPRICE	SALEDATE	QUANTITY
1	Cat	450.09	2018-05-29	9
2	Dog	666.66	2018-06-01	3

4. Next, modify the **SELECT** query as follows:

```
select count(*) from PETSHOP
```

Once you have completed this step, you should see all five rows of the PETSHOP table.

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Database: PETSHOP

```
SELECT count(*) FROM PETSHOP;
```

Tip: Autocomplete with Ctrl+Enter or Cmd+Enter

Submit query

Results

All commands ran successfully

```
SELECT count(*) FROM PETSHOP
```

count(*)
5

5. You have successfully created and loaded the **PETSHOP** table.

Exercise 2: Create and load data in the table using an SQL script file

In this exercise, you will learn how to create and load data into a table by running a script containing the CREATE and INSERT SQL commands.

1. Download the script file to your computer:

- [BookShop-CREATE-INSERT.sql](#)
- Copy the contents of the script file and paste it in the datasette text area
- Select Submit query

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Practice SQL

Database: PETSHOP

```
-- Drop the tables in case they exist

DROP TABLE IF EXISTS BookShop;
DROP TABLE IF EXISTS BookShop_AuthorDetails;

-- Create the table
```

Tip: Autocomplete with Ctrl+Enter or Cmd+Enter

[Submit query](#)

Results

All commands ran successfully

3. Next, click the **home** link at the top of the page.

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home PESHOP

Practice SQL

Database: PESHOP

```
-- Drop the tables in case they exist

DROP TABLE IF EXISTS BookShop;
DROP TABLE IF EXISTS BookShop_AuthorDetails;

-- Create the table
```

Tip: Autocomplete with Ctrl+Enter or Cmd+Enter

Submit query

Results

All commands ran successfully

4. This step will redirect you to a page displaying **Databases and Tables**.

- Select the **BookShop** table under the **PetShop** database.

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Datasette

[_memory](#)

0 tables

[_internal](#)

92 rows in 5 tables

[columns](#), [foreign_keys](#), [tables](#), [indexes](#), [databases](#)

[Instructors](#)

12 rows in 3 tables

[BookShop](#), [BookShop_AuthorDetails](#), [Instructor](#)

[PETCHOP](#)

10 rows in 2 tables

[PETCHOP](#), [BookShop](#)

Powered by Datasette

5. You will be able to view the **columns** and **data** of the **Bookshop** table.

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home / PETSHOP



BookShop

5 rows

- column -

=

Apply

 [View and edit SQL](#)

This data as [json](#), [CSV \(advanced\)](#)

Suggested facets: [AUTHOR_NAME](#), [AUTHOR_BIO](#), [AUTHOR_ID](#), [PUBLICATION_DATE](#) (date)

Show charting options

Link	rowid ▼ ⚙	BOOK_ID ⚙	TITLE ⚙	AUTHOR_NAME ⚙	AUTHOR_BIO ⚙	AUTHOR_ID ⚙
1	1	B101	Introduction to Algorithms	Thomas H. Cormen	Thomas H. Cormen is the co-author of Introduction to Algorithms, along with Charles Leiserson, Ron Rivest, and Cliff Stein. He is a Full Professor of computer science at Dartmouth College and currently Chair of the Dartmouth College Writing Program.	123
2	2	B201	Structure and Interpretation of Computer Programs	Harold Abelson	Harold Abelson, Ph.D., is Class of 1922 Professor of Computer Science and Engineering in the Department of Electrical Engineering and Computer Science at MIT and a fellow of the IEEE.	456

Congratulations! You have completed this lab and are ready for the next topic.

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